Hyunah Ko

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Education

- **KAIST** — Seoul, South Korea
  M.S. student, Kim Jaechul Graduate School of AI — *Current*
  CVLAB; Advisor: Prof. Seungryong Kim

- **Yonsei University** — Seoul, South Korea
  B.S. in Civil and Environmental Engineering
  Double Major: Computer Science — *GPA: 3.95/4.5*

Research Interests

My research focuses on geometry-aware visual representations for 3D/4D understanding and robotics. I am particularly interested in lifting 2D foundation-model features into dense 3D representations while preserving semantic information and consistency across views. I also study compact 3D/4D reconstruction and efficient vision-language-action models, with the long-term goal of enabling robots to perceive and interact with the physical world.

- Geometry-guided multi-view representations, dense correspondence, and 3D feature lifting
- Compact 3D/4D representations and feed-forward scene reconstruction
- Visual representations and efficient vision-language-action models for robotics

Publications and Preprints

- **One Loss Is Not Enough: Cross-Objective Structured Pruning for Reasoning Vision–Language–Action Models**
  Seongchan Kim, Soowon Son, Sangoh Kim, Hyunah Ko, Eunbeen Hong, Hyunkoo Lee, Sanghyun Lee, WonJun Moon, Kihong Kim, Dongyeon Son, Woongchan Nam, Seungryong Kim
  NeurIPS Workshops, 2026. Grounded and Faithful Vision-Language Models for Real-World Deployment; On-Device Intelligence: Foundation Models under Real-World Constraints; PTA: From Pretrained Representations to Acting Agents.

- **C4G: Learning Global Motion with Compact Gaussians for Feed-Forward 4D Reconstruction**
  Mungyeom Kim*, Minkyeong Jeon*, Honggyu An*, Jaewoo Jung, Hyuna Ko, Jisang Han, Hyeonseo Yu, Donghwan Shin, Sunghwan Hong, Takuya Narihira, Kazumi Fukuda, Yuki Mitsufuji, Seungryong Kim
  arXiv preprint, 2026.
  Paper Project

- **C3G: Learning Compact 3D Representations with 2K Gaussians**
  Honggyu An*, Jaewoo Jung*, Mungyeom Kim, Chaehyun Kim, Minkyeong Jeon, Jisang Han, Kazumi Fukuda, Takuya Narihira, Hyuna Ko, Junsu Kim, Sunghwan Hong, Yuki Mitsufuji, Seungryong Kim
  CVPR, 2026. Also at the Workshop on Open-World 3D Scene Understanding and Representations, CVPR 2026.
  Paper Project

- **Backbone Augmented Training for Adaptations**
  Jae Wan Park, Junhyeok Kim, Youngjun Jun, Hyunah Ko, Seong Jae Hwang
  AAAI Workshop on Artificial Intelligence with Biased or Scarce Data, 2026 • Oral. arXiv preprint, 2025.
  Paper

- **Probing and Leveraging Video Diffusion Transformer Features for Robust Point Tracking**
  Soowon Son, Honggyu An, Jisu Nam, Hyunah Ko, Chaehyun Kim, Dahyun Chung, Siyoon Jin, Jung Yi, Junhwa Hur, Seungryong Kim
  arXiv preprint, 2025; revised 2026. DiTracker
  Paper Project

- **Fourier Decomposition for Explicit Representation of 3D Point Cloud Attributes**
  Donghyun Kim, Chanyoung Kim, Hyunah Ko, Seong Jae Hwang
  arXiv preprint, 2025; revised 2026.
  Paper

* Equal contribution. Also published as Hyuna Ko.

Research Experience

- **CVLAB, KAIST** Jul. 2025 – Present
  M.S. student; previously undergraduate researcher. Advisor: Prof. Seungryong Kim
- **Medical Imaging and Computer Vision Lab, Yonsei University**
  Undergraduate Researcher; Advisor: Prof. Seong Jae Hwang Jul. 2024 – Jul. 2025
- **Environmental Biotechnology & Thermal Laboratory, Yonsei University**
  Undergraduate Researcher; Advisor: Prof. Joonhong Park Jul. 2023 – Sep. 2023

Team Projects

- **Startup Investment Round Prediction via BM-based Clustering**
  May. 2025 – Sep. 2025
  Predicted investment rounds (Seed–IPO) using multi-source time-series data. Validated BM clusters via statistical metrics (Cohesion/Separation) and identified stage-specific growth factors.
- **Single Image to 3D Generation via Video Diffusion and Gaussian Splatting**
  Apr. 2025 – May. 2025
  Developed a unified framework integrating **CameraCtrl** and **3D Gaussian Splatting** to ensure temporal and geometric consistency. Optimized 3D mesh generation via SDS loss, achieving high-fidelity results; awarded **3rd Place at YAICON 6th**.
- **Recipe Recommendation via Object Detection and Knowledge Graph QA***
  Oct. 2024 – Nov. 2024
  Built a recipe recommendation system using YOLOv5 and knowledge-augmented prompts with Stable Diffusion image generation.
- **Pet Body Measurement via Depth Estimation**
  Oct. 2024 – Nov. 2024
  Developed a depth-based system for pet body length measurement using MMPose and fine-tuned Depth Anything v2 with patch-wise refinement.
- **Generative Pet Memorials with LLaMA3 and Stable Diffusion**
  Aug. 2024 – Sep. 2024
  Generated personalized pet letters and images using LoRA-finetuned LLaMA3 and Stable Diffusion with ControlNet and style transfer.
- **LLM-based Personalized Health Report Generation**
  Jul. 2024 – Sep. 2024
  Developed a personalized report generation system using medical LLMs and Retrieval-Augmented Generation.
- **Seoul Night Bus Route Analysis and Design***
  Jul. 2024 – Aug. 2024
  Analyzed public transportation and mobility data to identify underserved regions in Seoul. Applied EDA and regression modeling to propose new night bus routes based on demand patterns.
- **Room Generation with Stable Diffusion and ControlNet***
  Apr. 2024 – May. 2024
  Trained a customized model using ControlNet and Stable Diffusion to generate personalized room images from segmentation and depth data.

- **A(uto)ssignment: Automated Problem Extraction from Textbooks**
  Mar. 2024 – Apr. 2024
  Built an automated pipeline using YOLOv7 and PORORO OCR to detect and extract questions from textbook PDFs, enabling custom problem set generation.

- **Traffic Commute Analysis for Northern & Southern Seoul**
  Jan. 04, 2024 – Jan. 30, 2024
  Analyzed ridership and station data to identify high-congestion areas in commutes from Gyeonggi Province to Seoul using EDA and statistical methods.

* Served as a team leader.

Selected Coursework

- **AI/ML**: Introduction to AI, NLP and ChatGPT, Large Language Model, Computer Vision, Reinforcement Learning
- **Systems**: Operating System, Data Structures, OOP, Multicore and GPU Programming
- **Math/Stats**: Linear Algebra, Introduction to Statistics, Probability and Statistics

Skills

- **Programming Languages**: Python (Expert), Java (Advanced), C++ (Intermediate), JavaScript (Beginner)
- **Development Tools**: Git, Docker, Linux/Unix, Matlab
- **Languages**: Korean (Native), English (Proficient)

Honors and Awards

- **Grand Prize, 2024 Jeju Satellite Data Utilization Competition**
  Jul. 2024 – Aug. 2024
  Proposed a data-driven solution and business model to address regional issues in Jeju Island by analyzing satellite imagery and identifying actionable insights.

- **77th out of 514, AirQo African Air Quality Prediction Challenge, Zindi**
  Mar. 2024 – Jun. 2024
  Developed a model to estimate air pollution levels using satellite imagery, addressing the high cost of ground-based sensor networks in large regions like Africa.

Extracurricular Activities

- **Yonsei Computer Club**
  Vice President (Feb. 2024 – Feb. 2025)
- **Yonsei Artificial Intelligence**
  Vice President (2025)
- **Yonsei Data Science Lab** *Dec. 2023 – Nov. 2024*